Shahar Eliyahu

B.Sc. Software Engineer

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SUMMARY

Software Engineering B.Sc. graduate with a major in Artificial Intelligence and a foundation in machine learning, computer vision, NLP and software development. Hands-on academic experience in Python, Java, Kotlin and C/C++, building Android applications, deep learning models and real-time data pipelines. Experienced in data annotation and dataset curation, model training, fine-tuning and evaluation, and in developing and debugging end-to-end applications. Analytical, detail-oriented, and motivated to further develop my skills in AI and software engineering.

PROJECTS & AWARDS

EasyFill · Final Project

- Built an Android application (**Kotlin**, **Jetpack Compose**) that helps people coping with PTSD fill out complex bureaucratic forms through intelligent form filling and adaptive accessibility support.
- Integrated **Azure AI Document Intelligence** via **Google Cloud Run** for PDF data extraction and automatic form completion.
- Developed a **multimodal distress-detection** system combining voice, facial signals, motion sensors and user interaction data, with **personalized baselines** that adapt the interface in real time.

Object Detection Pipeline · Computer Vision Project

- Built an end-to-end **object detection** pipeline for person and vehicle detection, using **Florence-2** for automatic **data annotation**, followed by dataset curation and **data augmentation**.
- Fine-tuned and compared several **YOLO** models with **transfer learning**, evaluated them with **mAP**, precision and recall, and selected a lightweight model for **Edge AI** deployment.

Tactical Combat AI Simulation · AI for Games Project

- Developed a real-time tactical combat simulation in **C++**, in which two AI teams (commander, warriors, medic, supply) fight autonomously.
- Implemented **A* pathfinding**, **BFS**, line-of-sight and **finite state machines (FSM)** for agent decision-making, within a modular **object-oriented design** with a class hierarchy for agent roles.

Real-Time Retail Trend Detection · Big Data Project

- Built a real-time e-commerce streaming pipeline using **Apache Kafka** and **Spark Structured Streaming**, with **stateful stream processing** and a **Random Forest** model to detect emerging product trends.

TestGuru · College Hackathon · 2nd Place

- Built an exam-practice platform that maps syllabus topics to past-exam questions and tracks performance by topic to identify weak areas.

EDUCATION

Afeka Academic College of Engineering · B.Sc. in Software Engineering, AI Major | 2026
Relevant Coursework: Computer Vision (100) · Machine Learning (98) · Big Data Analytics (97) · Natural Language Processing (95) · Object-Oriented Programming (100) · Computer Embedded Systems (97) · Compilation (96) · Data Structures (94)

SKILLS

Programming: Python · Java · Kotlin · C · C++ · SQL (PostgreSQL, MySQL) · MongoDB · REST APIs · OOP · Data Structures & Algorithms
Mobile: Android · Jetpack Compose
Machine Learning & Deep Learning: PyTorch · TensorFlow/Keras · scikit-learn · Hugging Face · CNN · Transfer Learning · Fine-tuning · Model Evaluation · Error Analysis
Computer Vision: OpenCV · YOLO (Ultralytics) · Florence-2 · MediaPipe · Object Detection · Image Segmentation · Data Annotation & Labeling · Data Augmentation · Edge AI
NLP: Transformers · DistilBERT · spaCy · NLTK · Word2Vec · LSTM
Data Engineering: Pandas · NumPy · Apache Spark (Structured Streaming) · Apache Kafka · EDA · Data Cleaning
Parallel Computing: OpenMP · MPI · Multithreading
Cloud & Services: Microsoft Azure · Azure AI Document Intelligence · Google Cloud Run · Firebase
AI-Assisted Development: Claude Code · Claude · ChatGPT · Cursor · GitHub Copilot
Tools: Git · GitHub · Linux · CUDA / GPU Training · Android Studio · Visual Studio

EXPERIENCE

Landa Digital Printing · Systems Technician | 2021-2024

- Assembled and calibrated complex digital printing systems following strict technical procedures and quality standards, and troubleshot hardware faults in a multidisciplinary engineering environment.

Basketball Coach | 2019-2021

- Coached children's basketball groups, planned training sessions and game preparation, and developed teamwork and discipline.

Military Service · Security Officer, Active Athlete | 2017-2019

- Responsible for facility security and security procedures, working under pressure with multiple teams.

LANGUAGES

Hebrew: Native | **English:** Fluent (technical reading and writing)